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INDIA'S ENERGY RESILIENCE COMMAND CENTER

FINAL PROJECT DOCUMENT

Sanjiv

A command center for the moment a supply shock stops being a headline and becomes a decision. It shows what is known, what is assumed, what happens next, and who must approve the response.

SCAN FOR REPOSITORY

GitHub repository
Demo folder
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WHY SANJIV EXISTS

Energy resilience is not one prediction. It is a chain of imperfect signals, physical constraints and human responsibility. I built Sanjiv to keep that chain visible - especially the uncertainty.

Built and documented by Yarra Vivek

Submission snapshot | 2026-07-22

Commit
`de5140a4d25aa88fac001e47030fd0b11d6f4258`

All screenshots and measured values are traceable to repository artifacts. PortWatch is a live-fetched daily observation; fixture and replay evidence is never presented as live vessel tracking.

02 / PROBLEM

A disruption does not arrive with clean data

The team still has to answer: What changed? What breaks next? What can we do? Which numbers are trustworthy? Who signs off?

IMAGINE THE MOMENT

The Strait of Hormuz loses capacity. Vessel feeds may be incomplete. Inventory may be unknown. A media spike may be wrong. The response team still needs a defensible next move:

+ What do we know right now?
+ What is inferred, modeled or assumed?
+ What happens if no one acts?
+ Which alternatives survive hard constraints?
+ Who is accountable for the final decision?

THE LINE I WOULD NOT CROSS

Sanjiv recommends; a human decides. It can observe, freeze a scenario, simulate consequences, compare candidate plans, audit the evidence and preserve a decision record.

It cannot place an order, charter a tanker, release a reserve, control a pipeline or operate a refinery. Public capacity metadata does not establish current fill, cargo ownership or commercial availability. That boundary is deliberate. [1][2]

PRIMARY USERS

01

Maritime and geopolitical analyst

Vessel and chokepoint activity, source health, risk explanation, provenance and historical comparison.

02

Refinery procurement head

Candidate suppliers, grades, landed cost, arrival timing, compatibility, sanctions and corridor risk.

03

Strategic reserve planner

Capacity versus opening-fill truth, draw schedule, logistics, policy floors and replenishment constraints.

04

Reviewer and approver

Evidence coverage, recomputation, rejected alternatives, immutable comments and explicit authority.

SCOPE DELIVERED IN THE REPOSITORY

Commodity	Operating mode	Decision outputs	Excluded claims
Crude oil; typed LPG extension	Live-fetched PortWatch observation plus checksummed replay and fixtures	No-action simulation; procurement; reserve; audit; approval; replay export	No live AIS claim; no private inventory; no confirmed cargo or contract; no execution

03 / SOLUTION

Follow one decision from signal to sign-off

Every handoff leaves a fingerprint. If evidence disappears along the way, the decision metric disappears with it.

THE SIX HANDOFFS

1 OBSERVE — Source mode, freshness and evidence →

2 FREEZE — Scenario plus immutable twin fingerprint →

3 SIMULATE — Paired baseline and no-action impact →

4 OPTIMISE — Three procurement and four reserve profiles →

5 AUDIT — Coverage, claims, hashes and recomputation →

6 APPROVE — Server-owned immutable human record

WHAT SITS BEHIND THE SCREENS

COMMAND CENTER	Next.js / React / typed REST and WebSocket clients
DOMAIN CORE	FastAPI modular monolith: sources, maritime, twin, scenario, simulation, optimisation, risk, audit, replay
WORKERS	Ingestion, scheduled refresh and compute workers import the same domain services
STATE	PostgreSQL + PostGIS + TimescaleDB (authoritative) | Redis (cache/fan-out) | MinIO (immutable artifacts)

I kept the first deployment deliberately compact: one measured modular monolith with shared domain services. Multiple API replicas come only after Redis WebSocket fan-out and deployment identity are in place.

FIVE WORDS WITH STRICT MEANINGS

- **OBSERVED** — Directly retrieved or supplied; normalization only.
- **DERIVED** — Deterministic calculation from identified inputs.
- **INFERRED** — Heuristic or probabilistic classification.
- **MODELED** — Simulator or optimiser output.
- **ASSUMPTION** — Visible editable input used when verification is absent.

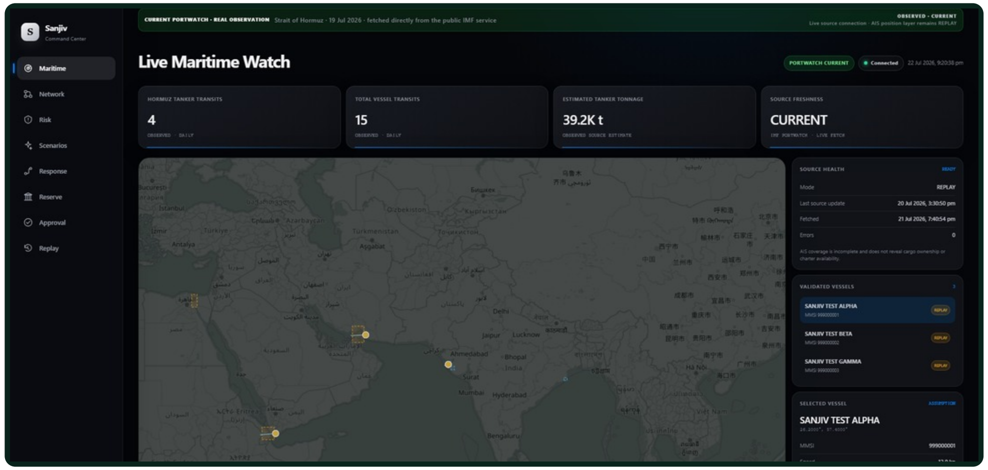
Required fields: source references; effective, fetch and compute timestamps; freshness; confidence; evidence IDs; transformation; units; model and contract versions.

04 / PRODUCT 1 OF 3

Real observations and replay stay visibly separate

PortWatch is live-fetched public data marked OBSERVED and CURRENT. The AIS position layer and modeled risk remain separately labeled replay or fixture evidence.

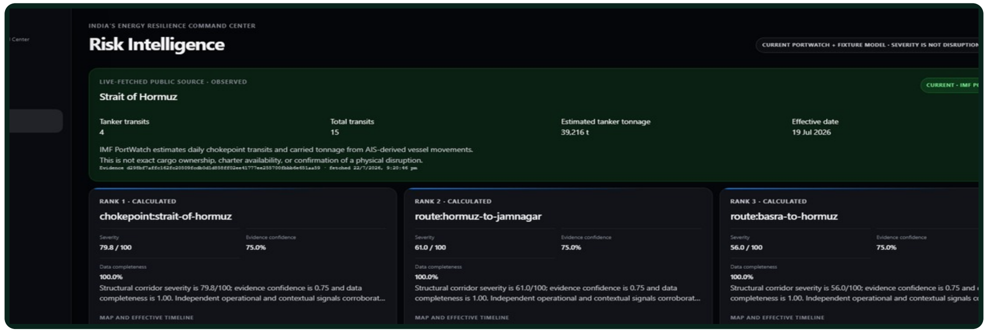
OBSERVE - COMMAND CENTER



The public IMF PortWatch service returned 4 tanker transits, 15 total transits and 39,216 t estimated tanker tonnage for 2026-07-19. The fetch is OBSERVED / CURRENT; it is a daily AIS-derived estimate, not live vessel tracking.

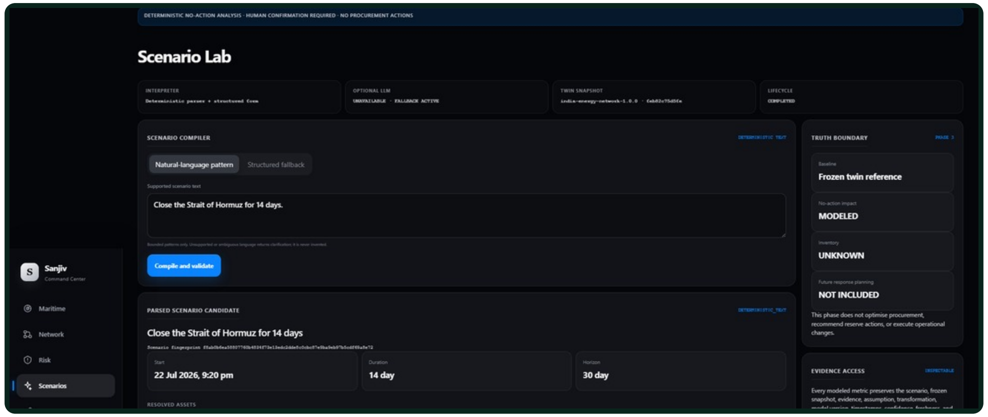
Source plane	Truth shown in the application
PortWatch daily passage	LIVE fetch; OBSERVED; CURRENT; source date and fetch time visible
AIS vessel positions	REPLAY; synthetic vessels; reported and inferred fields remain distinct
Risk model	FIXTURE / replay inputs remain separate from the observed PortWatch card

RISK INTELLIGENCE - OBSERVED SIGNAL



The observed PortWatch card sits above the structural risk ranking. The model remains explicitly fixture-based and severity is not presented as disruption probability.

SIMULATE - SCENARIO LAB



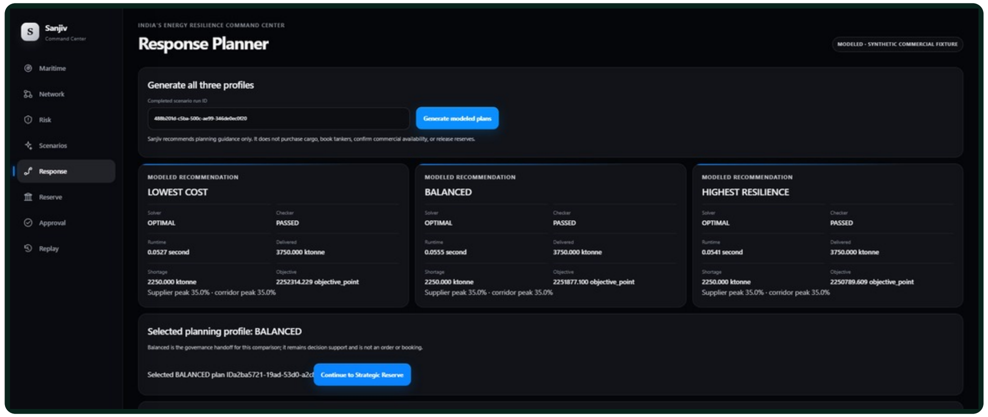
A person confirms the parsed disruption before the twin is frozen. Baseline and no-action outputs share one fingerprint; missing inventory remains UNKNOWN.

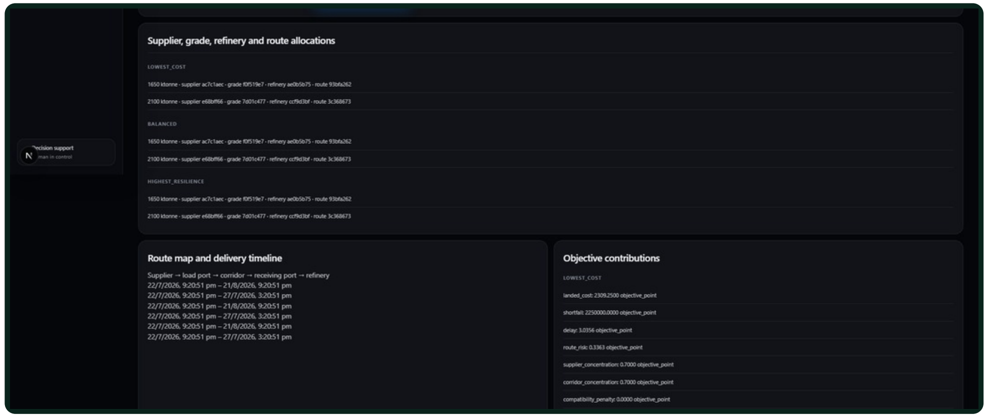
05 / PRODUCT 2 OF 3

Give the human real choices, not one magic answer.

Each candidate exposes the trade-off it is making. Public reserve capacity never quietly becomes an invented opening fill.

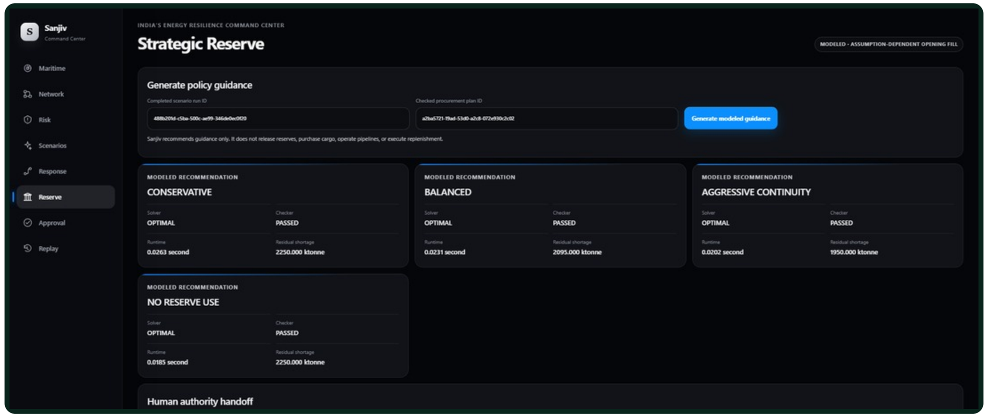
OPTIMISE - RESPONSE PLANNER

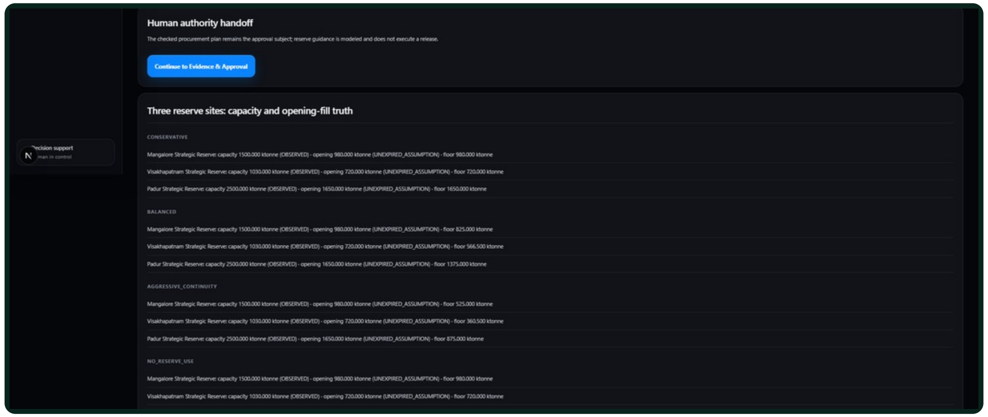




Builder's note. LOWEST COST, BALANCED and HIGHEST RESILIENCE share the same hard constraints and input fingerprint. The point is not a magic answer; it is a legible choice with checker state, shortage, cost components, rejections, evidence and assumptions.

RESERVE - STRATEGIC RESERVE





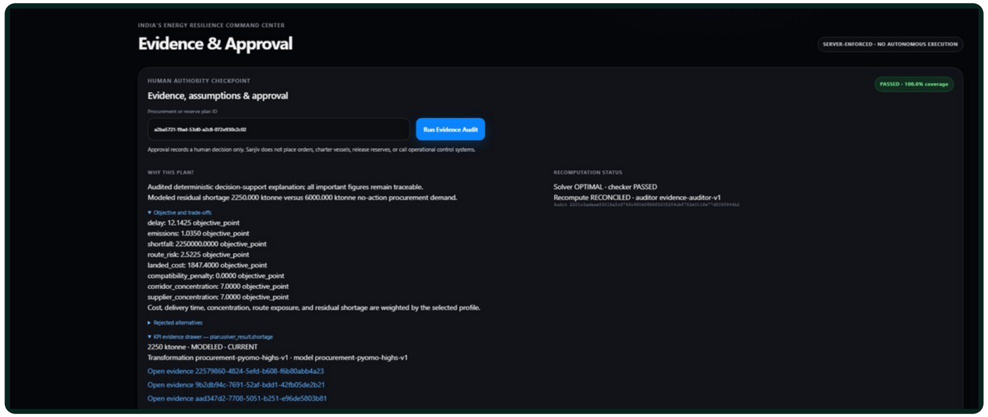
Builder's note. Capacity is not fill. The screen keeps public site capacity separate from the synthetic opening-fill assumption, then checks floors, draw and receipt limits, transit and residual shortage. No replenishment input means no hidden replenishment.

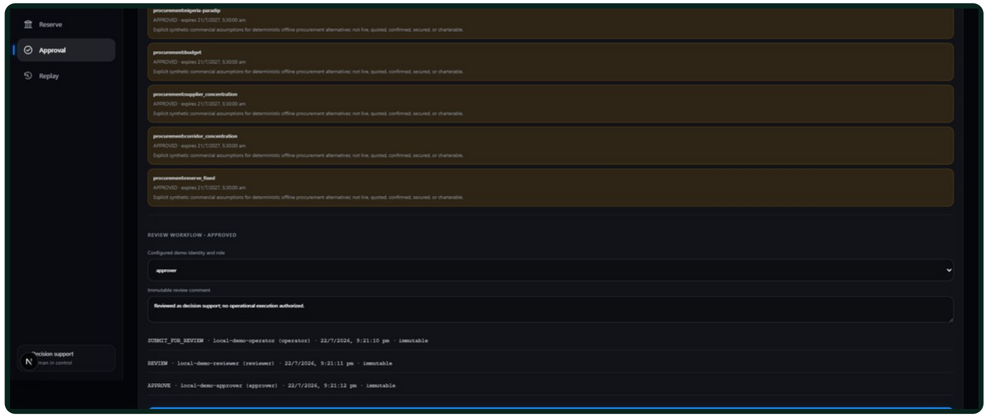
06 / PRODUCT 3 OF 3

A plan is not trusted because the interface says 'optimal'.

Trust comes from the evidence, the arithmetic and the person willing to sign their name to the decision.

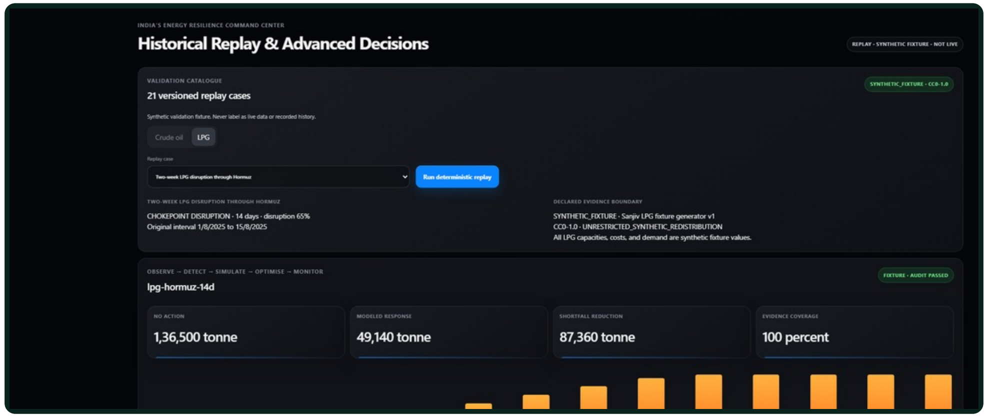
AUDIT AND APPROVE - EVIDENCE WORKSPACE

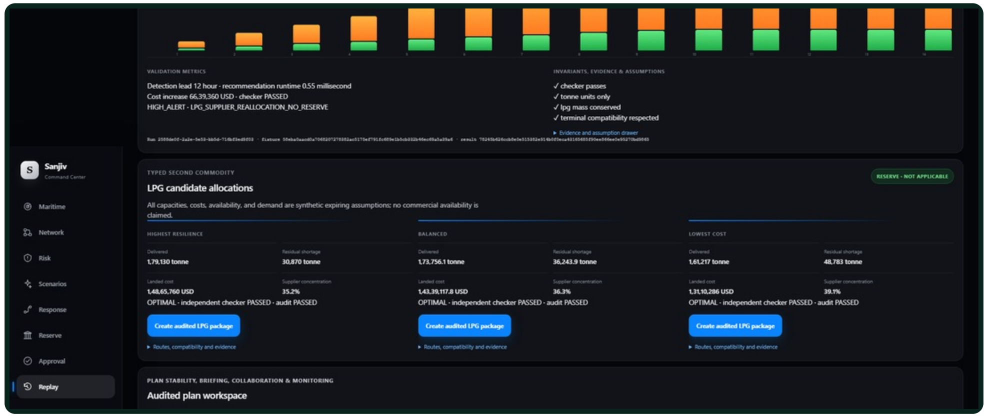




Builder's note. The captured audit is PASSED with 100.0% coverage and RECOMPUTE_RECONCILED. The approval stores server-resolved actor, role, UTC time and an immutable comment. It records a human decision; it never executes one.

MONITOR - HISTORICAL REPLAY AND LPG





Builder's note. Twenty-one versioned CC0 synthetic cases keep regression and demonstration repeatable. Classification, generator, license, assumptions, invariants and audit state stay visible. Seeded sensitivity is deterministic analysis, not a probability forecast.

07 / GOVERNANCE

Provenance must survive a hard question

If someone asks, 'Where did that number come from?', the answer should be one click away - or the number should not be on the decision screen.

EVIDENCE CHAIN

RAW	Hash and licensing permit
NORMALIZE	Schema, units, time, identity
METRIC	Truth class and full envelope
AUDIT	Freshness, claims, recompute
DECISION	Submit, review, approve/reject

Missing mandatory evidence blocks the metric from decision UI and briefing output.

SECURITY AND AUTHORITY BOUNDARY

+ Credentials stay server-side; browser configuration contains only the public API origin.
+ Production API and governance fail closed without configured server-side identities.
+ Origin, request size/type, rate, SSRF-safe adapter and bounded solver controls are implemented.
+ Independent checkers and the Evidence Auditor block failed plans, approvals and exports.
+ Terminal plans, audits, approvals, exports and snapshots are append-only or database-immutable.

Residual boundary: deployment IdP, TLS termination, multi-user session policy and Redis fan-out are operator-owned prerequisites.

REGISTERED EXTERNAL SOURCES AND EXACT TRUTH TREATMENT

Source	Repository use	Truth and limitation	Ref.
PPAC	Typed offline import boundary for refinery and energy statistics	Bundled operating values are assumptions; no PPAC payload is bundled	[1]
ISPRL	Public reserve-site capacity and connectivity metadata	Capacity may be observed; current fill stays unknown without verified input	[2]
IMF PortWatch	Live-fetched daily passage observation and baseline	Observed public estimate; current by source cadence; not live AIS	[3]
OFAC	Official sanctions list files	List entry observed; exact match derived; fuzzy match inferred	[4]
EIA / FRED	Optional price, supply and macro series	Series cadence, units, rights and attribution remain source-specific	[5][6]
NASA FIRMS / GDELT	Optional thermal and media signals	Signal is not proof of closure, attack, cause or damage	[7][8]

08 / VALIDATION

Here is what actually passed - and what it does not prove

Machine, dataset, timestamp and commit matter. Current tests were rerun; performance evidence was regenerated from a clean committed application source.

CURRENT NPM TEST
212 passed
188 Python + 23 web + 1 contracts

FAILED / SKIPPED
0 / 0
Exit 0; elapsed 200.5 s

BROWSER E2E COVERAGE
2 journeys
All 8 product screens; Observe to Monitor

REPLAY CATALOGUE
21 cases
Checksummed CC0 synthetic fixture

LOCAL FIXTURE PERFORMANCE (MILLISECONDS)

Metric	Median	p95	n
AIS ingest to map	24.296	24.296	1
Risk scoring	47.758	58.768	5
Scenario compilation	7.983	13.711	5
Simulation	55.625	77.405	5
Procurement optimisation	330.683	493.029	5
Reserve optimisation	176.739	214.952	5
Evidence audit	45.007	62.722	5
Briefing export	125.802	309.542	5
Signal to recommendation	775.790	980.187	5

Report classification: MEASURED_LOCAL_SYNTHETIC_FIXTURE. Five samples unless shown otherwise. These are release measurements, not production SLAs or model-accuracy claims.

BENCHMARK IDENTITY

Run `95dce13d-f2fc-49ad-bda6-07aa232f1545`
Report commit `de5140a4d25aa88fac001e47030fd0b11d6f4258`
Source state CLEAN_COMMITTED; timestamp 2026-07-22T16:06:57.578634Z

BROWSER, LOAD AND RESYNC

Map 35.40 FPS; interaction 412.407 ms; frame-time p95 139.9 ms across 120 frames.
Load: 64/64 successful at concurrency 8; p95 408.518 ms. Resync observed after 100 publishes; p95 2.774 ms.

DEPENDENCY FAILURE AND RECOVERY DRILL (MS)

Component	Detection	Recovery	Result
Redis	1030.028	30.933	PASS
MinIO	1028.974	1487.670	PASS
PostgreSQL	1028.329	1035.884	PASS
Compute worker	22427.376	1045.361	PASS
API	n/a	4754.407	PASS

Security: PASS (8 checks; 0 failure names). Backup/restore: PASS; 2,673,646 bytes; migration 20260721_0009; corrupt artifact rejected.

09 / PATH TO USE

The value is a better decision record

The repository proves observed-source ingestion, workflow mechanics and deterministic checks on fixtures. Operational value still requires licensed history and verified operator inputs.

DECISION VALUE NOW

+ One immutable chain from source state to approval.
+ Paired no-action and response alternatives on the same frozen inputs.
+ Independent feasibility checks and named rejected options.
+ Visible assumption ownership, expiry, freshness and evidence coverage.
+ Audited JSON/PDF packages plus monitored replay outcomes.

No rupee savings, avoided-loss value or decision-time reduction was measured.

VERIFIED DEPLOYMENT SHAPE

Docker Compose runs the same web/API images, three workers, PostgreSQL with PostGIS and TimescaleDB, Redis and MinIO for local and demo use.

+ Liveness and dependency-aware readiness
+ Source, worker and request-runtime status
+ Structured logs and trace-compatible identifiers
+ Backup/restore and single-dependency drills

SCALING POLICY

+ Scale vertically first using measured queue and request latency.
+ Add ingestion/refresh/compute worker replicas by workload.
+ Add Redis WebSocket fan-out before multiple API replicas.
+ Use managed database/object storage, external TLS and an operator collector.
+ Require an ADR and workload evidence before Kubernetes or domain microservices.

OPERATIONAL ADOPTION ROADMAP

01
Replace fixtures
Licensed recorded history and verified private/operator inputs.

02
Calibrate
Out-of-sample thresholds, model weights and decision policies with domain owners.

03
Secure identities
Deployment IdP, TLS edge, rotation, sessions and least privilege.

04
Scale and recover
Redis fan-out, managed HA, encrypted off-site backup and multi-region drills.

05
Accept and govern
Operator acceptance, data rights, model monitoring and change control.

METRICS FOR A REAL DEPLOYMENT - NOT RESULTS IN THIS SUBMISSION

Shortfall reduction; incremental cost per avoided shortage unit; inventory-cover extension; refinery-utilisation recovery; supplier and corridor concentration; decision time saved. Each requires a baseline, verified input data and an agreed measurement window.

10 / LIMITS AND REFERENCES

What I refused to fake

A convincing demo should make its limits easier to see, not easier to miss. Real PortWatch observations and fixture-based decision evidence remain explicitly separated.

CURRENT LIMITATIONS

- PortWatch passage data is live-fetched and labeled OBSERVED / CURRENT; AIS positions, decision-model inputs and the replay catalogue remain fixture or replay evidence where shown.
- Commercial price, supplier availability, port/refinery values and reserve opening fill are synthetic assumptions unless replaced by verified operator records.
- No order placement, tanker charter, reserve release, pipeline or refinery-control integration exists.
- Risk weights, alert thresholds and deterministic sensitivity are not calibrated disruption probabilities.
- The benchmark is a one-machine local synthetic-fixture run from clean application commit de5140a4d25a; it is release evidence, not a production SLA or model-accuracy result.
- The recovery drill stops one dependency at a time on a local single-host stack; it does not establish multi-region failover.
- Production requires deployment IdP, TLS, key rotation, sessions, managed backups and Redis fan-out before horizontal API scaling.
- Licensed recorded validation, domain-owner calibration and operator acceptance remain required before operational decision use.

AUTHORITATIVE REFERENCES

[1] PPAC - Installed Refining Capacity
https://ppac.gov.in/infrastructure/installed-refinery-capacity

[2] ISPRL - Strategic reserve facilities
https://isprlindia.com/aboutus.asp

[3] IMF PortWatch - Data and methodology
https://portwatch.imf.org/pages/data-and-methodology/

[4] OFAC - Sanctions List Service
https://ofac.treasury.gov/sanctions-list-service

[5] U.S. EIA - API v2 documentation
https://www.eia.gov/opendata/documentation.php

[6] FRED - API overview
https://fred.stlouisfed.org/docs/api/fred/overview.html

[7] NASA FIRMS - Area API
https://firms.modaps.eosdis.nasa.gov/api/area/

[8] GDELT - DOC 2.0 API
https://blog.gdeltproject.org/gdelt-doc-2-0-api-debuts/

SUBMISSION LINKS

Repository QR

Repository
https://github.com/vivekyarra/Sanjiv

Demo
https://drive.google.com/drive/folders/1JDFeLyQOd4g74BhE63Yuh6OCGmCoaf6K?usp=sharing

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Application commit
`de5140a4d25aa88fac001e47030fd0b11d6f4258`